Low Level Design

on

Healthcare Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Introduction

The Healthcare planning Assistant is an Agentic AI-based system designed to help users generate a complete healthcare plan based on their medical condition and location. The system uses a Planner Agent that intelligently breaks down the user's request into smaller tasks and coordinates multiple specialized agents, such as a Research Agent, Hospital Finder Agent, and Scheduler Agent.

1.1. Scope of the document

This document defines the Low-Level Design (LLD) of the Healthcare planning Assistant Agent.
It covers:

- Internal system architecture
- Component-level design
- Data and process flow
- APIs and class structures

The goal is to provide a detailed technical blueprint for developers to implement the system.

1.2. Intended Audience

The intended audience for the Healthcare Planning Assistant includes patients, caregivers, and general users seeking medical guidance. This system helps them by providing a structured healthcare plan based on their disease and location, reducing the effort of searching for reliable treatment options and nearby hospitals. It is especially useful for individuals who want quick, organized, and personalized healthcare information, enabling them to make better decisions about their treatment and manage their health more effectively.

1.3. System overview

The Healthcare Planning Assistant is an Agentic AI system that:

- Accepts user inputs (disease, location)
- Breaks down healthcare tasks using a Planner Agent
- Uses multiple agents:
  - Research Agent → finds treatment info
  - Hospital Finder Agent → finds hospitals
  - Scheduler Agent → creates timeline
- Produces a complete healthcare plan

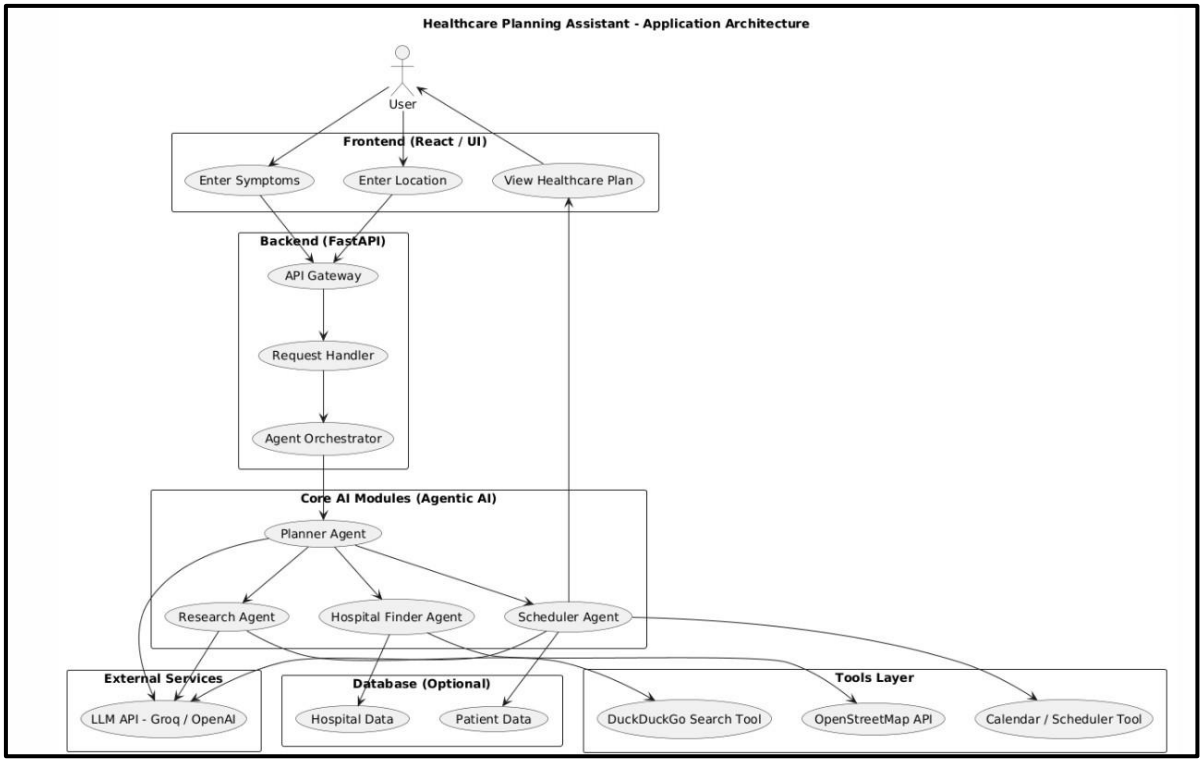
System Design

2.1. Application Design

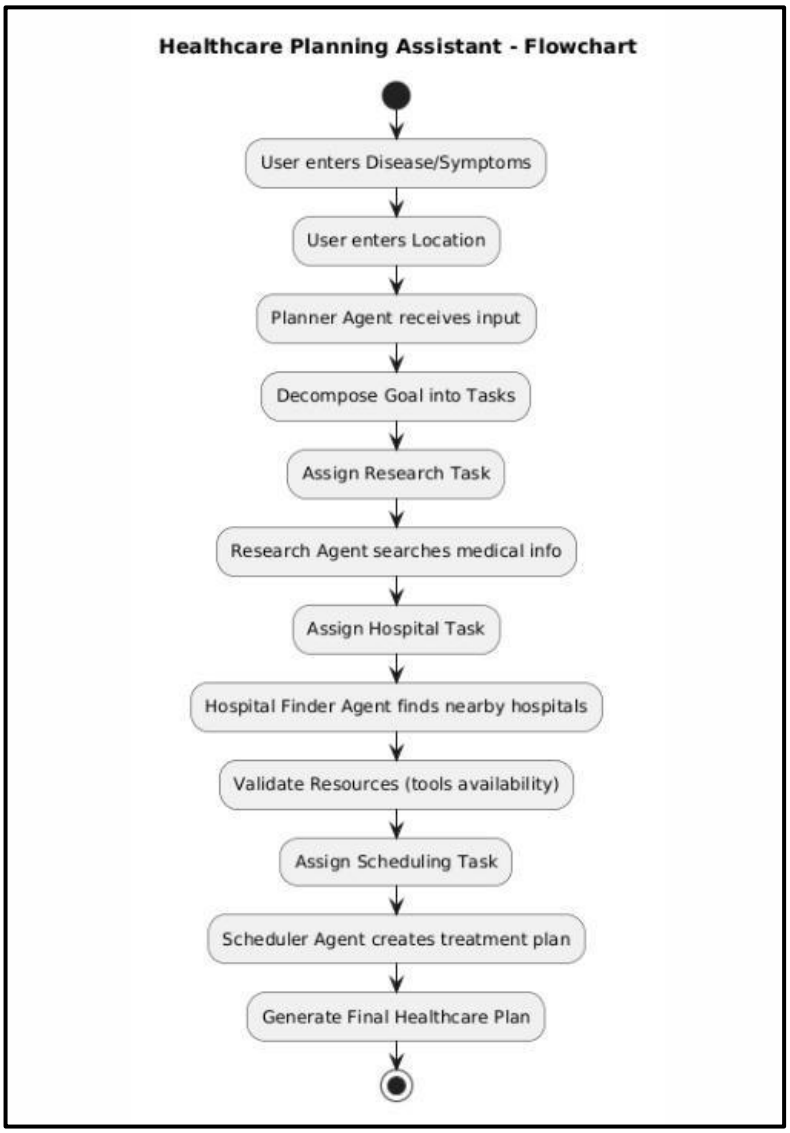
Architecture Type: Modular Multi-Agent Architecture

Layers:

- Frontend (UI) → React (User input/output)
- Backend (Flask) → API + Agent orchestration
- Agent Layer → Planner + Sub-agents
- Data Layer → Database (SQLite)



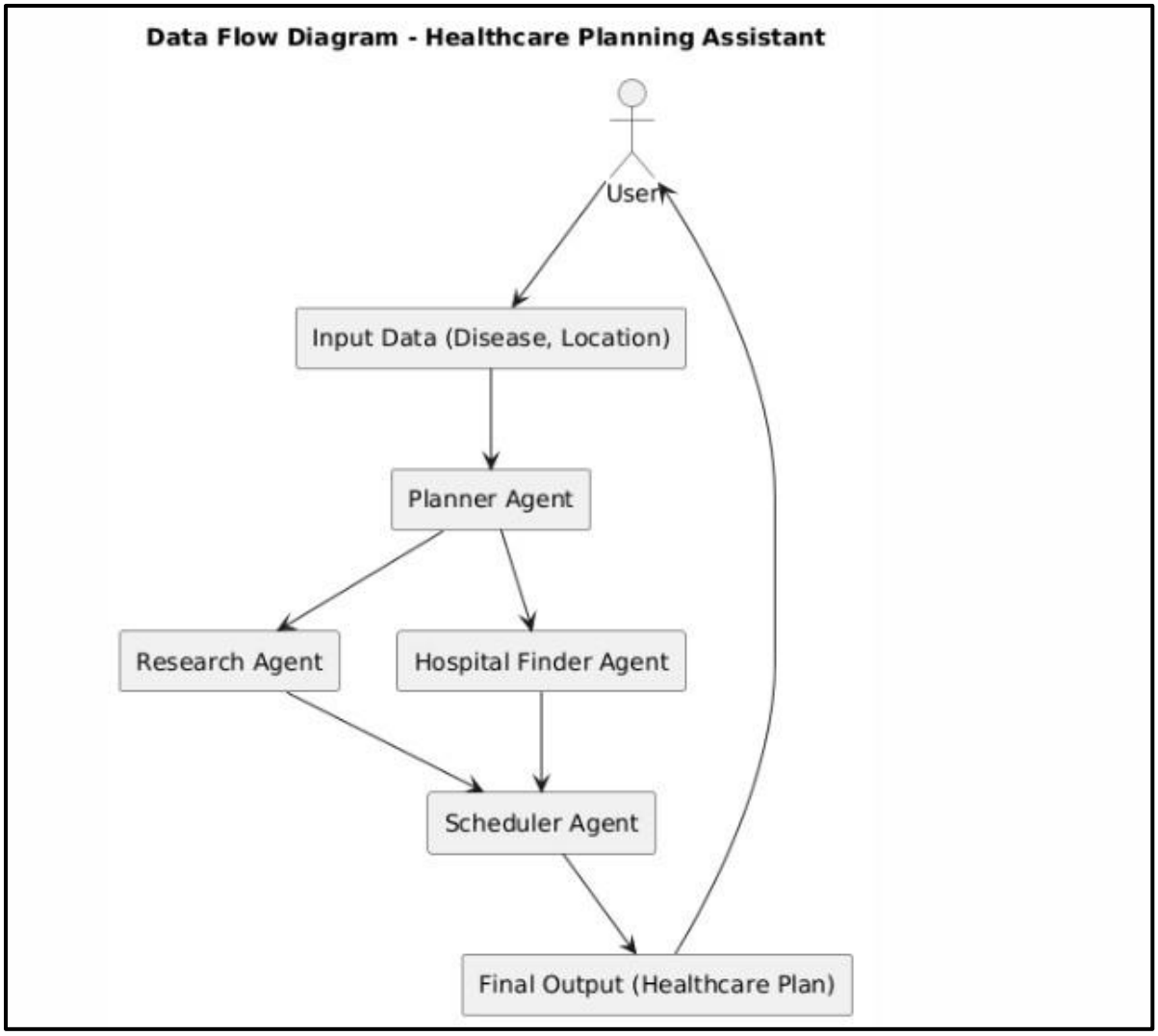
2.2. Process Flow



starts with user input, then the planner agent decomposes the goal into tasks, assigns them to specialized agents, and finally combines their outputs to generate a complete healthcare plan.

2.3. Information Flow

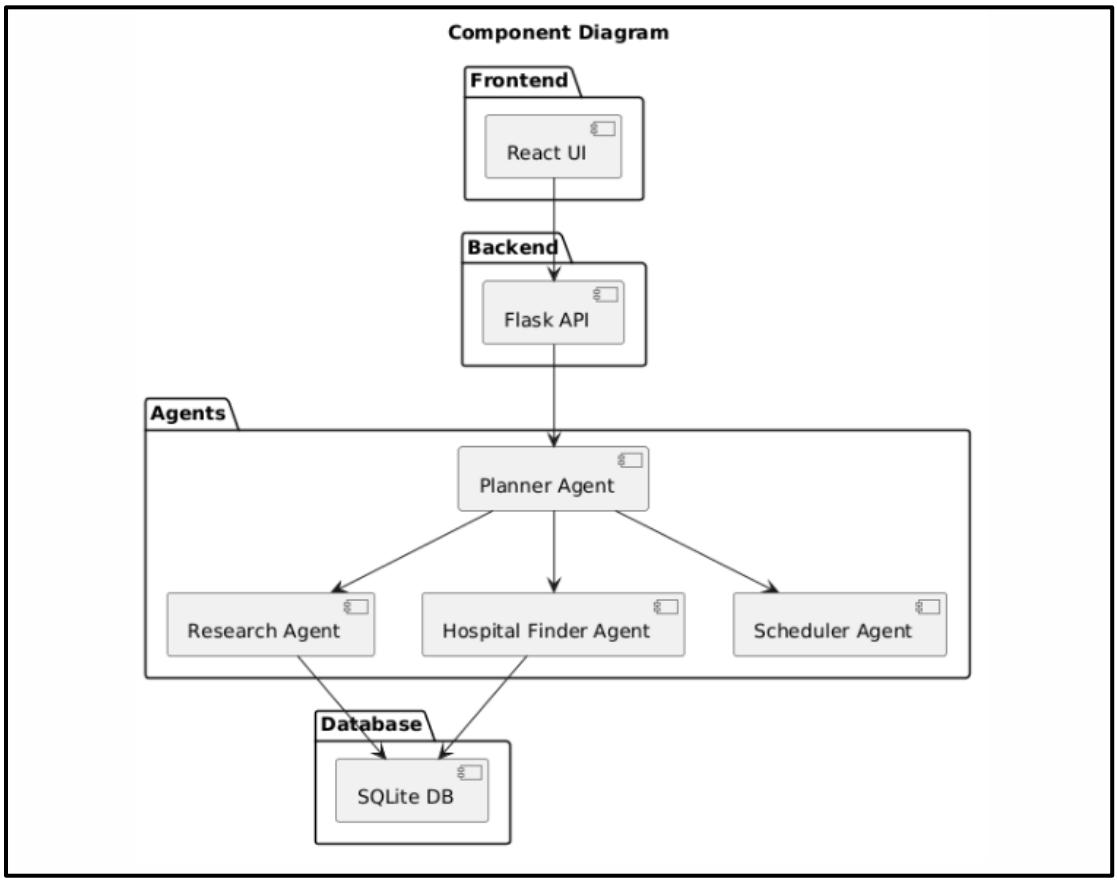
- Input → Disease, Location
- Intermediate → Treatment data, hospital list, schedule
- Output → Structured healthcare plan



2.4. Components Design

Components:

- Frontend UI
- API Controller
- Planner Agent (Core Brain)
- Research Agent
- Hospital Finder Agent
- Scheduler Agent
- Database



2.5. Key Design Considerations

- Scalability → Add more agents easily
- Modularity → Independent agent design
- Fault Tolerance → Failure in one agent shouldn't break system
- Extensibility → Add features like insurance, telemedicine
- Performance → Parallel execution of agents
- Security → Protect health data

2.6. API Catalogue

1. Get Healthcare Plan

POST /generate-plan

Request:

```
{
 "disease": "diabetes",
"location": "Indore"
}
```

Response:

```
{
"treatment": "...",
"hospitals": [...],
"schedule": [...]
 }
```

2. Get Hospitals

GET /hospitals?location=Indore

3. Get Treatment Info

GET /treatment?disease=diabetes

References

- Agentic AI Concepts
- LangChain Documentation
- Flask Documentation
- React Documentation
- SQLite Docs